Linear algebra – 9

Matrices over $\mathbb{Z}$ and over a finite field

Homework

12 сентября 2026 года

Theoretical minimum. $GL_n(\mathbb{Z})$ and unimodular matrices; the Smith normal form and the index of a sublattice; $|GL_n(\mathbb{F}_q)|$ and Gaussian binomial coefficients; reduction modulo a prime; the Cauchy–Binet formula and the matrix-tree theorem; Frobenius' theorem on the eigenvalues of an integer matrix of finite order.

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1. Compute $|GL_n(\mathbb{F}_q)|$ and the number of k-dimensional subspaces of $\mathbb{F}_q^n$. Let X be the number of fixed vectors of a uniformly random element of $GL_n(\mathbb{F}_q)$. Prove that for $n \geqslant k$ the moment $\mathsf{E}X^k$ does not depend on n and equals the total number of subspaces of $\mathbb{F}_q^k$; in particular, find $\mathsf{E}X$ and the variance of X for $n \geqslant 2$.

2. (ФКН, осень 2021) Let A be a 2×2 matrix with rational entries and suppose that $A^n = -E$ for some positive integer n. Prove that $A^2 = -E$ or $A^3 = -E$.

3. (SEEMOUS 2009) Let $SL_2(\mathbb{Z})$ be the set of integer 2×2 matrices of determinant 1.

   a) Find $A, B, C \in SL_2(\mathbb{Z})$ with $A^2 + B^2 = C^2$.

   b) Show that there are no $A, B, C \in SL_2(\mathbb{Z})$ with $A^4 + B^4 = C^4$.

4. (VJIMC 2018, cat. I) Find all primes p such that p^3 divides the determinant of the $(p+6)\times(p+6)$ matrix whose diagonal entries are $2^2, 3^2, 4^2, \ldots, (p+7)^2$ and all of whose off-diagonal entries are equal to 1.

5. (Romania national olympiad 1997) Let $m \geqslant 2$, $n \geqslant 1$ and let $A = (a_{ij})$ be an integer $n \times n$ matrix. Prove that for every permutation $\sigma \in S_n$

there is a function $\varepsilon\colon \{1,\dots,n\} \to \{0,1\}$ such that replacing the entries $a_{\sigma(1)1},\dots,a_{\sigma(n)n}$ by $a_{\sigma(i)i} + \varepsilon(i)$ yields a matrix whose determinant is not divisible by m.

6. (IMC) For a prime p let $GL_2(\mathbb{Z}/p\mathbb{Z})$ be the group of invertible 2×2 matrices over $\mathbb{Z}/p\mathbb{Z}$ and let S_p be the symmetric group on p elements. Show that there is no injective group homomorphism $\varphi\colon GL_2(\mathbb{Z}/p\mathbb{Z}) \to S_p$.

7. (IMC) Ivan writes the matrix $\begin{pmatrix} 2 & 3 \\ 2 & 4 \end{pmatrix}$ on the board and repeats the following operation: he chooses a row or a column and multiplies or divides it entry-wise by the other row or column respectively. Can he end up with $\begin{pmatrix} 2 & 4 \\ 2 & 3 \end{pmatrix}$?

8. (Мехмат, 2015) Find all primes p for which the following is true: if a 2015×2015 matrix over the field with p elements has the property that its rank drops whenever an arbitrary nonzero entry is replaced by zero, then every row contains at most one nonzero entry.

9. (SEEMOUS 2012) Let $A = (a_{ij})$ be the $n \times n$ matrix in which a_{ij} is the remainder of $i^j + j^i$ upon division by 3. Find the greatest n for which $\det A \neq 0$.

10. (Kirchhoff) Prove the matrix-tree theorem: the number of spanning trees of a graph equals any cofactor of its Laplacian. Then compute the Smith normal form of the Laplacian of K_n and interpret the answer as the structure of the critical group.